Prospects for Further Cooperation

Several potential avenues for continued cooperation are intended or may be pursued following the completion of the planned project:

1. **Applications for new joint projects.** The partners intend to apply for new research projects, including similar bilateral initiatives, in the research area addressed by the present proposal. This is a highly active field, with new theoretical and practical challenges emerging and being addressed on a regular basis. It is therefore very reasonable to pursue further joint research projects after the completion of the proposed activities.
2. **Workshops.** The partners intend to co-organize at least one workshop after the end of this project associated with a fitting international conference, e.g. GECCO. This activity would promote the research field and attract other researchers at an international level.
3. **Cooperation in master's-level education.** The Mathematics and Informatics study programme is currently preparing to launch its first cohort of a modern master's programme in computer science. Collaboration at the master's level, including teaching contributions from academics at both institutions, could enrich the educational experience of students. Given the strong overlap between the partners' research interests—particularly discrete optimization, algorithms, and graph theory—such cooperation would be especially appropriate. It could also involve joint supervision of master's theses, participation in collaborative projects, and opportunities for students to benefit from the expertise of both institutions.
4. **Institutional partnerships and student mobility.** The partners intend to strengthen links between their universities through joint applications to programmes such as Erasmus+ and Horizon Europe. Such initiatives could enable master's students from both institutions to spend a period at the partner institution, attend courses relevant to their interests, complete examinations, and collect the corresponding credits recognized by their home institution.
5. **Invited research seminars.** Researchers from each institution shall be invited to present their work within the other institution's research seminar series also after this project. Both partners already organize seminars aimed at doctoral students, academic staff, and invited researchers. Reciprocal seminar visits would help strengthen institutional ties and stimulate new ideas for collaborative projects and joint scientific publications.
6. **Development of an international research network and COST Action.** In the near future, and following an expansion of the network of collaborating researchers, the institutions may bring together additional partners to establish an international research team. This would create a foundation for applying to relevant COST calls, potentially with a proposed COST Action focused on generative artificial intelligence in discrete optimization and the application of these methods to real-world problems involving large-scale networks.